Exponential sums associated to prehomogeneous vector spaces and their relatives

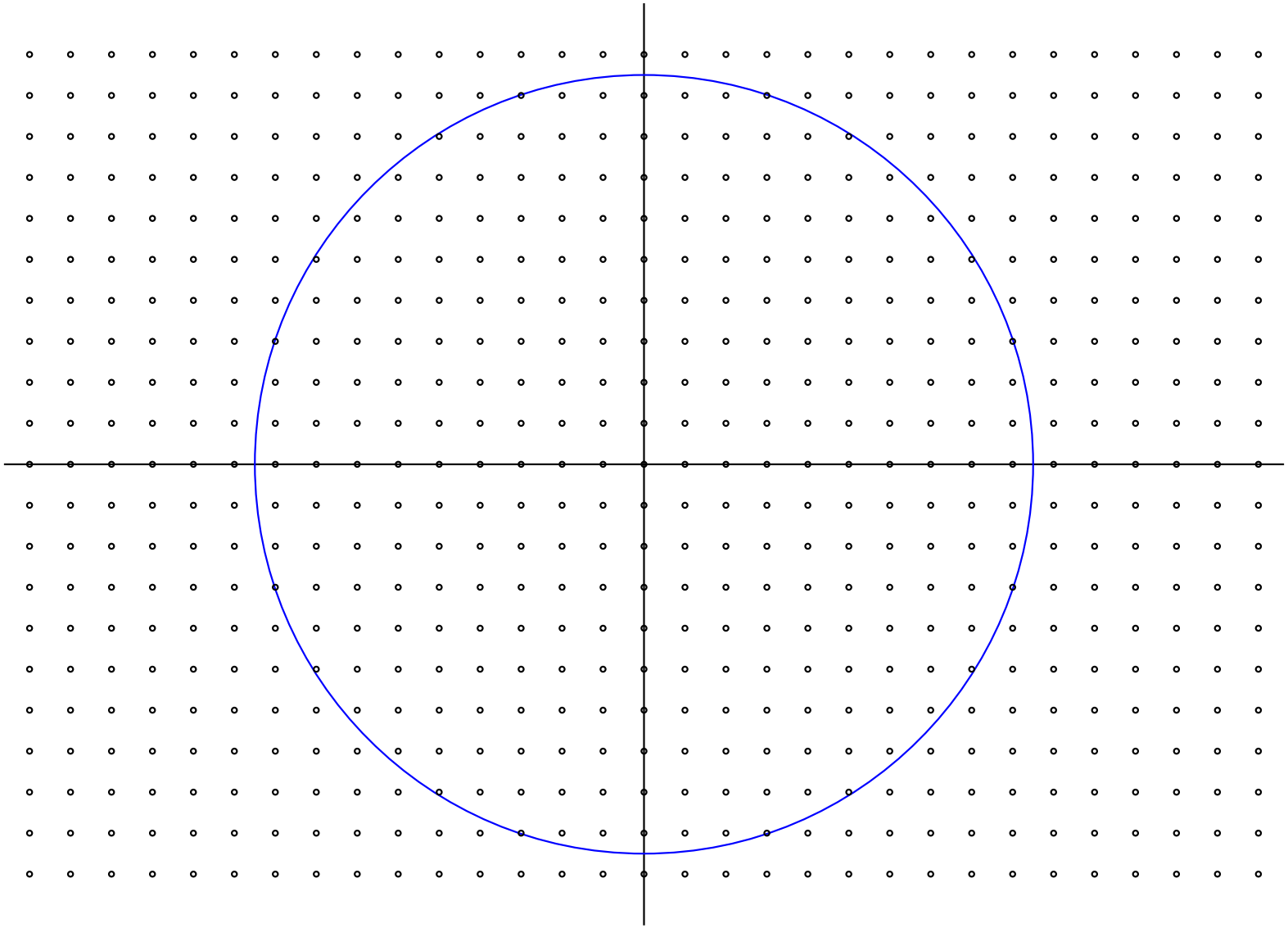
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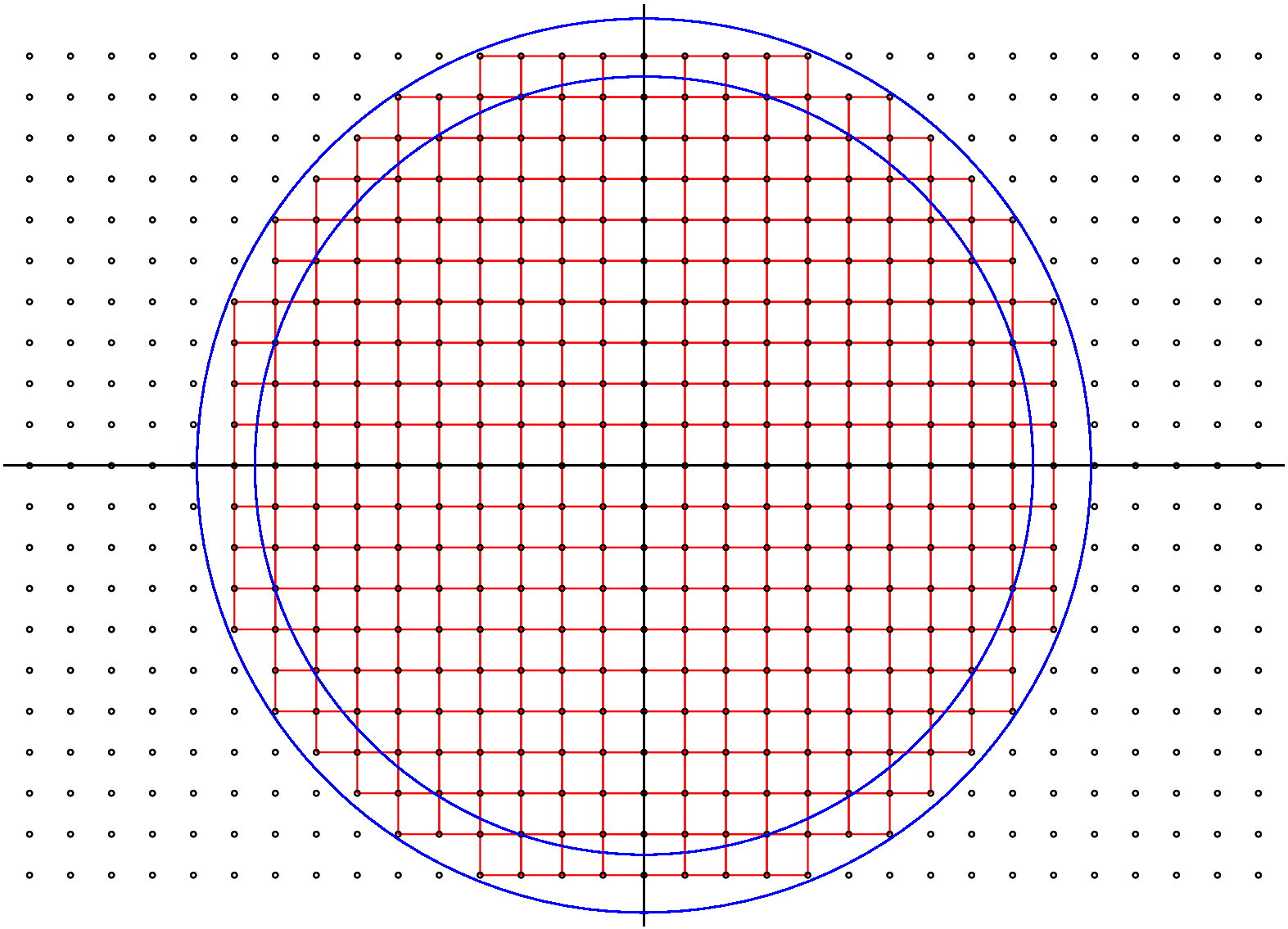
University of South Carolina, Columbia, SC, USA

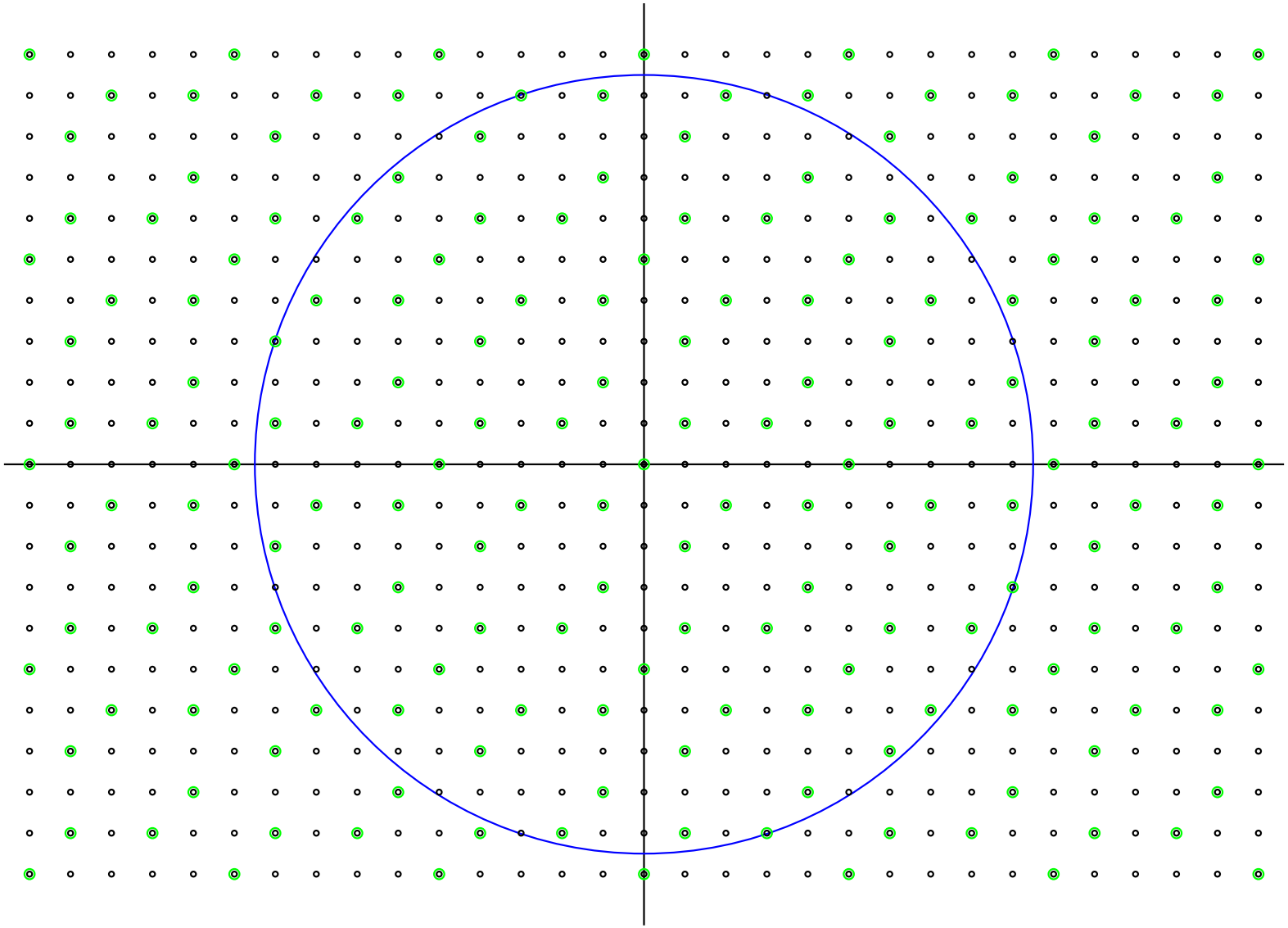
AMS Sectional Meeting, Kennesaw State University

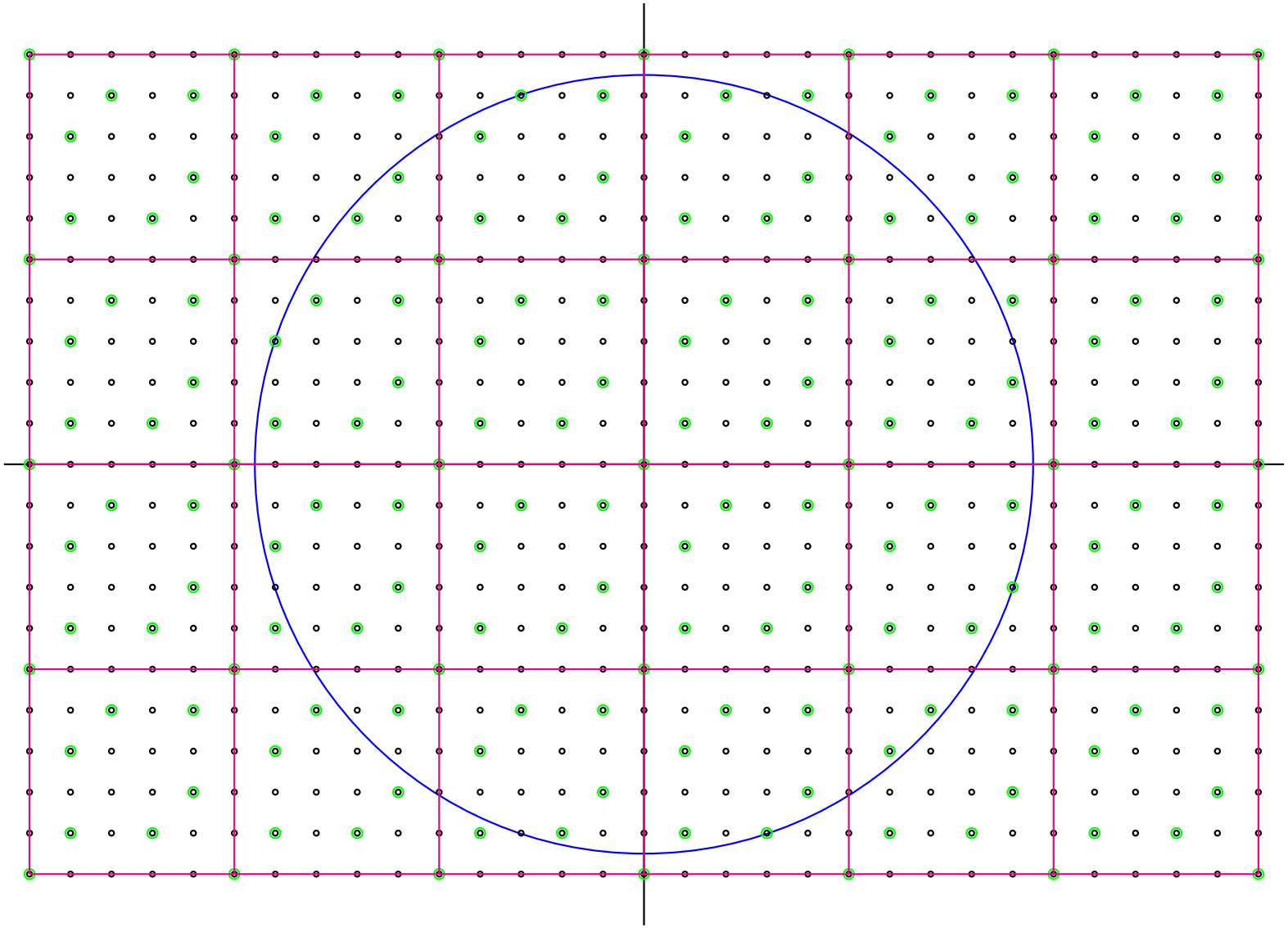
`thornef.github.io/exposums-kennesaw.pdf`











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- Exact formulas;
- L_1-norm bounds;
- 'Filtration' formulas;
- Whatever else we can prove.

Example: Pólya-Vinogradov

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Theorem (Pólya-Vinogradov inequality, special case)

Let χ be a primitive Dirichlet character $\pmod q$. *Then we have*

$$\left| \sum_{n=M+1}^{M+N} \chi(n) \right| < q^{\frac{1}{2}} \log q.$$

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Proof. By Fourier inversion, we have

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with $|\tau(\overline{\chi})| = q^{1/2}$, so that

$$\sum_{n=M+1}^{M+N} \chi(n) = \frac{1}{\tau(\overline{\chi})} \sum_{a=1}^{q} \overline{\chi}(a) \sum_{n=M+1}^{M+N} e^{2\pi i a n/q},$$

and the innermost sum is a geometric series.

A Bhargava-style metatheorem

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- *$G(\mathbb{Z})$-orbits on a lattice $V(\mathbb{Z})$; where G is an* **algebraic group** *acting (often* **prehomogeneously***) on a vector space V;*
- *Some nice class of arithmetic objects we want to count.*

Examples

- $V = \mathrm{Sym}^3(2)$, $G = \mathrm{GL}_2$: cubic rings; 3-torsion in class groups (Levi, Delone-Faddeev, Davenport-Heilbronn)
- $V = 2 \otimes \mathrm{Sym}^2(3)$, $G = \mathrm{GL}_2 \times \mathrm{GL}_3$.
  quartic rings; 2-torsion in class groups of cubic fields (Wright-Yukie; Bhargava)
- $V = 4 \otimes \wedge^2(5)$, $G = \mathrm{GL}_4 \times \mathrm{SL}_5$
  quintic rings (Wright-Yukie; Bhargava)
- $V = \mathrm{Sym}^4(2)$, $G = \mathrm{PGL}_2$
  2-Selmer elements of elliptic curves (Birch–Swinnerton-Dyer, Bhargava-Shankar)
- ... and more!
  (Bhargava, Ho, Shankar, Varma, X. Wang, Wood, ...)

An explicit evaluation

Theorem (Taniguchi-T., 2011)

We have

$$\widehat{\Phi_{p^2}}(v) = \begin{cases} p^{-2} + p^{-3} - p^{-5} & v/p : \text{of type } (0), \\ p^{-3} - p^{-5} & v/p : \text{of type } (1^3), (1^2 1), \\ -p^{-5} & v/p : \text{of type } (111), (21), (3). \\ p^{-3} - p^{-5} & v : \text{of type } (1^3_{**}), \\ -p^{-5} & v : \text{of type } (1^3_{*}), (1^3_{\max}), \\ 0 & \text{otherwise}, \end{cases}$$

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So:

$$\frac{1}{p^8} \sum_{v \in V(\mathbb{Z}/p^2\mathbb{Z})} |\widehat{\Phi_{p^2}}(v)| \ll p^{-7}.$$

Consequence – Improved Davenport-Heilbronn

Theorem (Bhargava-Taniguchi-T.)

We have

$$N_3(X) = CX + C'X^{\frac{5}{6}} + O(X^{\frac{3}{5}+\epsilon}) + O(X^{1-\frac{1}{8-7+2}+\epsilon}).$$

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where

- V is the lattice of binary cubic forms,
- Φ_{p^2} is the characteristic function of nonmaximal at p.

Singular binary cubic forms

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Theorem (Mori 2010)

We have

$$\widehat{\Phi_p}(v) = \begin{cases} p^{-1} + p^{-2} - p^{-3} & (v = 0), \\ p^{-2} - p^{-3} & (v \text{ has splitting type } (1^3) \text{ or } (1^2 1)), \\ -p^{-3} & (\text{otherwise}). \end{cases}$$

Pairs of Ternary Quadratic Forms

With $p \neq 2$, let Ψ_p be the characteristic function of singular pairs of ternary quadratic forms.

Theorem (Taniguchi-T., 2016)

$$\widehat{\Psi_p}(x) = \begin{cases} p^{-1} + 2p^{-2} - p^{-3} - 2p^{-4} - p^{-5} + 2p^{-6} + p^{-7} - p^{-8} & x \in \mathcal{O}_{\mathbf{0}}, \\ p^{-3} - p^{-4} - 2p^{-5} + 2p^{-6} + p^{-7} - p^{-8} & x \in \mathcal{O}_{D\mathbf{1}^2}, \\ 2p^{-4} - 5p^{-5} + 3p^{-6} + p^{-7} - p^{-8} & x \in \mathcal{O}_{D\mathbf{11}}, \\ p^{-4} - 3p^{-5} + 2p^{-6} + p^{-7} - p^{-8} & x \in \mathcal{O}_{C\mathrm{s}}, \\ -p^{-5} + p^{-6} + p^{-7} - p^{-8} & x \in \mathcal{O}_{D\mathbf{2}}, \mathcal{O}_{D\mathrm{ns}}, \mathcal{O}_{C\mathrm{ns}}, \mathcal{O}_{B\mathbf{11}}, \mathcal{O}_{B\mathbf{2}}, \\ -p^{-6} + 2p^{-7} - p^{-8} & x \in \mathcal{O}_{\mathbf{1}^2\mathbf{1}^2}, \\ p^{-6} - p^{-8} & x \in \mathcal{O}_{\mathbf{2}^2}, \\ p^{-7} - p^{-8} & x \in \mathcal{O}_{\mathbf{1}^4}, \mathcal{O}_{\mathbf{1}^3\mathbf{1}}, \mathcal{O}_{\mathbf{1}^2\mathbf{11}}, \mathcal{O}_{\mathbf{1}^2\mathbf{2}}, \\ -p^{-8} & x \in \mathcal{O}_{\mathbf{1111}}, \mathcal{O}_{\mathbf{112}}, \mathcal{O}_{\mathbf{22}}, \mathcal{O}_{\mathbf{13}}, \mathcal{O}_{\mathbf{4}}. \end{cases}$$

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Obtain Fourier transform formulas for all twenty orbits.

More Pairs of ternary quadratic forms (Hough, 2018)

Theorem 2. *The Fourier transform of the maximal set is supported on the mod p orbits* $\mathscr{O}_0, \mathscr{O}_{D1^2}, \mathscr{O}_{D11}$ *and* $\mathscr{O}_{D2}$. *It is given explicitly in the following tables.*

(1) Case $\mathscr{O}_0$, $\xi = p\xi_0$.

(6.1)

Orbit	$p^{-12}\widehat{\mathbf{1}_{\max}}(p\xi_0)$	*Orbit size*
$\mathscr{O}_0$	$(p-1)^4p(p+1)^2(p^5+2p^4+4p^3+4p^2+3p+1)$	1
$\mathscr{O}_{D1^2}$	$-(p-1)^3p(p+1)^4$	$(p-1)(p+1)(p^2+p+1)$
$\mathscr{O}_{D11}$	$-(p-1)^3p(2p^3+6p^2+4p+1)$	$(p-1)p(p+1)^2(p^2+p+1)/2$
$\mathscr{O}_{D2}$	$(p-1)^2p(2p^2+3p+1)$	$(p-1)^2p(p+1)(p^2+p+1)/2$
$\mathscr{O}_{Dns}$	$(p-1)^2p(2p^2+3p+1)$	$(p-1)^2p^2(p+1)(p^2+p+1)$
$\mathscr{O}_{Cs}$	$-p^7+5p^5-3p^4-3p^3+p^2+p$	$(p-1)^2p(p+1)^2(p^2+p+1)$
$\mathscr{O}_{Cns}$	$(p-1)^2p(2p^2+3p+1)$	$(p-1)^2p^3(p+1)(p^2+p+1)$
$\mathscr{O}_{B11}$	$(p-1)^2p(2p^2+3p+1)$	$(p-1)^2p^2(p+1)^2(p^2+p+1)/2$
$\mathscr{O}_{B2}$	$(p-1)^2p(2p^2+3p+1)$	$(p-1)^3p^2(p+1)(p^2+p+1)/2$
$\mathscr{O}_{1^4}$	$p(p^3-3p^2+p+1)$	$(p-1)^3p^2(p+1)^2(p^2+p+1)$
$\mathscr{O}_{1^31}$	$p(p^3-3p^2+p+1)$	$(p-1)^3p^3(p+1)^2(p^2+p+1)$
$\mathscr{O}_{1^21^2}$	$(p-1)^2p(3p+1)$	$(p-1)^2p^4(p+1)^2(p^2+p+1)/2$
$\mathscr{O}_{2^2}$	$-(p-1)p(p+1)^2$	$(p-1)^3p^4(p+1)(p^2+p+1)/2$
$\mathscr{O}_{1^211}$	$p(p^3-3p^2+p+1)$	$(p-1)^3p^4(p+1)^2(p^2+p+1)/2$
$\mathscr{O}_{1^22}$	$p(p^3-3p^2+p+1)$	$(p-1)^3p^4(p+1)^2(p^2+p+1)/2$
$\mathscr{O}_{1111}$	$-p^3+p^2+p$	$(p-1)^4p^4(p+1)^2(p^2+p+1)/24$
$\mathscr{O}_{112}$	$-p^3+p^2+p$	$(p-1)^4p^4(p+1)^2(p^2+p+1)/4$
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Binary Quartic Forms

Theorem (Ishitsuka, Taniguchi, T., Xiao)

For a prime $p > 3$, for singular binary quartic forms we have

$$\widehat{\Phi_p}(v) = \begin{cases} p^{-1} + p^{-2} - p^{-3} & (v = 0), \\ p^{-2} - p^{-3} & (v \text{ has splitting type } (1^4) \text{ or } (1^3 1)), \\ \chi_{12}(p)(p^{-4} - p^{-3}) & (v \text{ has splitting type } (1^2 1^2)), \\ \chi_{12}(p)(p^{-4} + p^{-3}) & (v \text{ has splitting type } (2^2)), \\ \chi_{12}(p)p^{-4} & (v \text{ has splitting type } (1^2 11) \text{ or } (1^2 2)), \\ \chi_3(p)\left(\frac{I(v)}{p}\right) \cdot p^{-4} & (J(v) = 0, I(v) \neq 0), \\ a(E'_v)p^{-4} & (J(v) \neq 0, \mathrm{Disc}(v) \neq 0). \end{cases}$$

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Here $a(E'_v) := p + 1 - \#E'_v(\mathbb{F}_p)$, where

$$E'_v \ : \ y^2 = x^3 - 3I(v)x^2 + J(v)^2.$$

Theorem (Wafik)

Now let Φ_p count doubly singular binary quintic forms, with $p > 5$. If $J(f) = K(f) = L(f) = 0$ then

$$p^5\widehat{\Phi_p}(f) = \begin{cases} 2p^3 - p & f = 0 \\ 2p^2 - 2p & f \in (1^5) \\ 2p^2 - 3p + 1 & f \in (1^4 1) \\ p^2 - 2p + 1 - (p-1)\left(\frac{6}{p}\right) & f \in (1^3 1^2) \\ p^2 - p + 2 + \left(\frac{6}{p}\right) & f \in (1^3 11) \\ p^2 - p + \left(\frac{6}{p}\right) & f \in (1^3 2), \end{cases}$$

and otherwise

$$p^5\widehat{\Phi_p}(f) = \begin{cases} -d_p(f)p + \omega_p(f) - 1 & L = K = 0 \neq J \\ (1 + s_p(f) - d_p(f))p + \omega_p(f) + a_p(f) - 1 & L = 0 \neq K \\ p\left[\left(\frac{L(f)}{p}\right) s_p(f) - d_p(f)\right] + \omega_p(f) + a_p(f) - 1 & L \neq 0. \end{cases}$$

Binary quintic forms (2)

Theorem (Ishitsuka)

Let Φ_p count all singular binary quintic forms, with $p > 5$. Write $f = (a_0, a_1, \ldots, a_5)$ and define the cubic covariant

$$\mathrm{Cat}(x, y; f) = \det\left(x \begin{pmatrix} a_1 & a_2 & a_3 \\ a_2 & a_3 & a_4 \\ a_3 & a_4 & a_5 \end{pmatrix} - y \begin{pmatrix} a_0 & a_1 & a_2 \\ a_1 & a_2 & a_3 \\ a_2 & a_3 & a_4 \end{pmatrix} \right).$$

If $\mathrm{Cat}(x, y; f) \neq 0$, then

$$\boxed{p^4 \widehat{\Phi_p}(f) = C(f), \qquad C(f) \in \{-4, -3, \ldots, 3, 4\}.}$$

Moreover, $C(f)$ has an explicit geometric description:

$$C(f) = - \sum_{\substack{[Q] \in \mathbb{P}^2(\mathbb{F}_p) \\ \langle f, xQ^2 \rangle = \langle f, yQ^2 \rangle = 0}} \left(\frac{\mathrm{Disc}(Q)}{p} \right).$$

Here Q ranges over binary quadratic forms up to scalars, and the sum has at most four terms. Every value from -4 to 4 occurs.

The Fouvry-Katz Theorem

Let Y be a (locally closed) subscheme of $\mathbb{A}^n_{\mathbb{Z}}$, of dimension d. Take $V = \mathbb{A}^n$, p prime, and Φ_p the the characteristic function of $Y(\mathbb{F}_p)$.

Theorem (Fouvry-Katz, 2001)

There exists a filtration of subschemes

$$\mathbb{A}^n_{\mathbb{Z}} \supseteq X_1 \supseteq \cdots \supseteq X_j \supseteq \cdots \supseteq X_n$$

with X_j of codimension j, so that

$$|\widehat{\Phi_p}(y)| \leq C p^{-n+\frac{d}{2}+\frac{j-1}{2}}$$

away from $X_j(\mathbb{F}_p)$.

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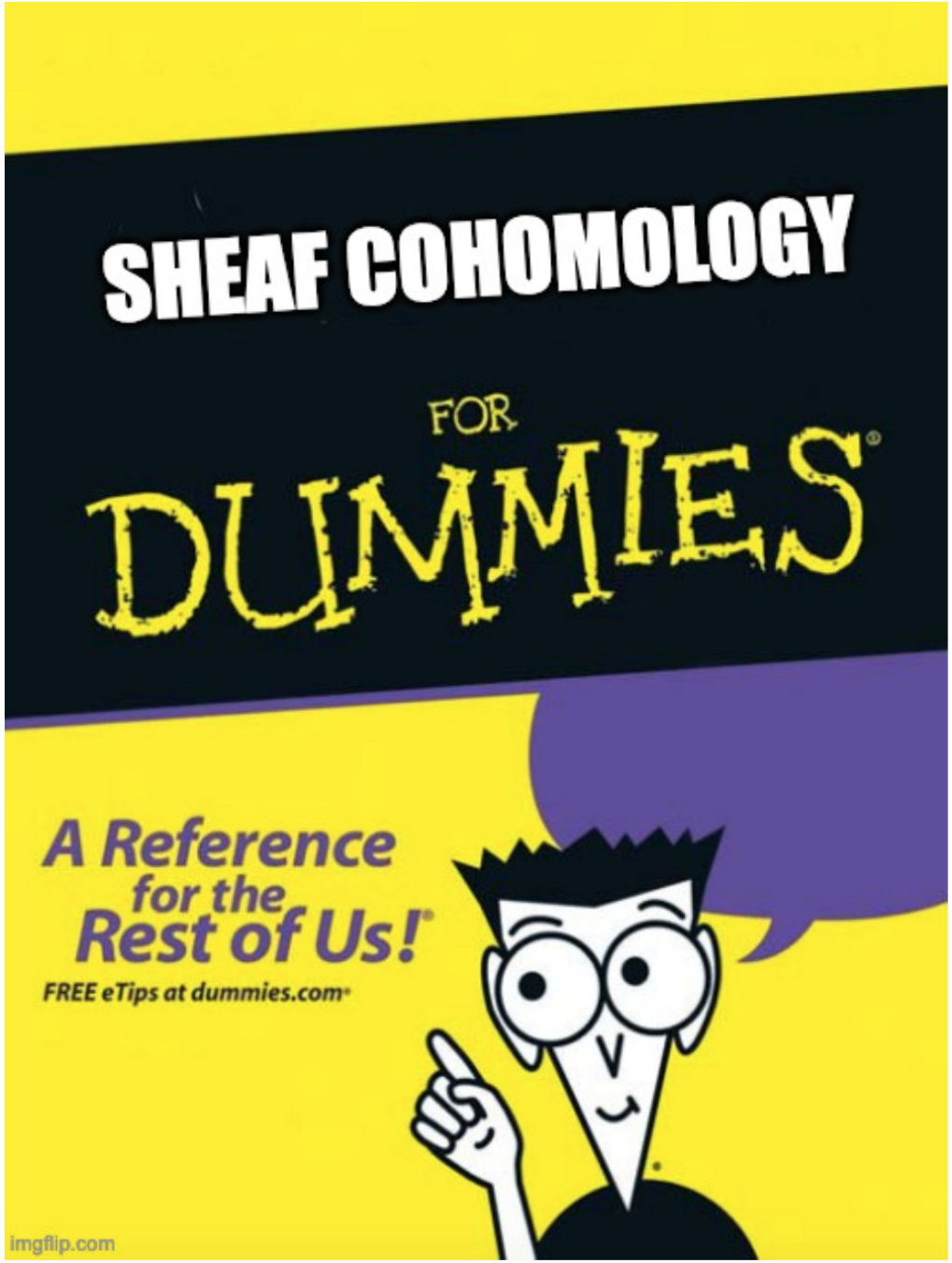
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  ..... T.-You?? .....

These questions are *way* too hard for LLMs. OpenAI, don't even *try*.


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The subspace method

Mori's theorem again:

- V be binary cubic forms;
- Φ_p is now the characteristic function of singular forms.

Theorem (Mori 2010)

We have

$$\widehat{\Phi_p}(v) = \begin{cases} p^{-1} + p^{-2} - p^{-3} & (v = 0), \\ p^{-2} - p^{-3} & (v \text{ has splitting type } (1^3) \text{ or } (1^2 1)), \\ -p^{-3} & (\text{otherwise}). \end{cases}$$

Proof of Mori's theorem

A binary cubic form v is singular if it has a repeated root.

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We obtain the decomposition

$$\Phi_p = \sum_{g \in \mathrm{GL}_2(\mathbb{F}_p)/B(\mathbb{F}_p)} \mathbf{1}_{[0,0,*,*]} - p\mathbf{1}_{[0,0,0,0]},$$

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with

$$\widehat{\mathbf{1}_{[0,0,*,*]}} = p^{-2}\mathbf{1}_{[*,*,0,0]}.$$

So,

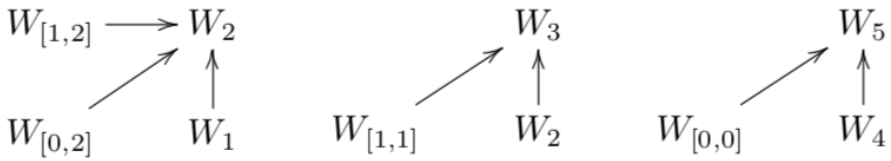
$$\widehat{\Phi_p} = p^{-2}\Phi_p - p^{-3} + p^{-1}\mathbf{1}_{[0,0,0,0]}.$$

Fun with pairs of ternary quadratic forms

(II) Secondly, again by the same method, we count for the following five subspaces

$$W_1 := \begin{bmatrix} 0&0&0&0&0&0 \\ 0&0&*&0&*&* \end{bmatrix}, \quad W_2 := \begin{bmatrix} 0&0&0&0&0&* \\ 0&0&*&0&*&* \end{bmatrix}, \quad W_3 := \begin{bmatrix} 0&0&*&0&*&* \\ 0&0&*&0&*&* \end{bmatrix} \quad \text{and} \quad W_4 := \begin{bmatrix} 0&0&0&0&0&0 \\ 0&0&0&*&0&* \end{bmatrix}, \quad W_5 := \begin{bmatrix} 0&0&0&*&0&* \\ 0&0&0&*&0&* \end{bmatrix}$$

with the following steps:



Our result is:

Subspace	$\mathcal{O}_0$	$\mathcal{O}_{D1^2}$	$\mathcal{O}_{D11}$	$\mathcal{O}_{C\mathrm{s}}$	$\mathcal{O}_{C\mathrm{ns}}$
$W_1 = \begin{bmatrix} 0&0&0&0&0&0 \\ 0&0&*&0&*&* \end{bmatrix}$	1	$q-1$	$q(q^2-1)$		
$W_2 = \begin{bmatrix} 0&0&0&0&0&* \\ 0&0&*&0&*&* \end{bmatrix}$	1	q^2-1	$q(q^2-1)$	$(q^2-1)(q^2-q)$	
$W_3 = \begin{bmatrix} 0&0&*&0&*&* \\ 0&0&*&0&*&* \end{bmatrix}$	1	q^2-1	$q(q^2-1)(q+1)$	$q(q^2-1)^2$	$q^2(q^2-1)(q^2-q)$

Subspace	$\mathcal{O}_0$	$\mathcal{O}_{D1^2}$	$\mathcal{O}_{D11}$	$\mathcal{O}_{D2}$	$\mathcal{O}_{B11}$
$W_4 = \begin{bmatrix} 0&0&0&0&0&0 \\ 0&0&0&*&0&* \end{bmatrix}$	1	$2(q-1)$	$\frac{1}{2}(q-1)^2$	$\frac{1}{2}(q-1)^2$	
$W_5 = \begin{bmatrix} 0&0&0&*&0&* \\ 0&0&0&*&0&* \end{bmatrix}$	1	$2(q^2-1)$	$\frac{1}{2}(q-1)(q^2-1)$	$\frac{1}{2}(q-1)(q^2-1)$	$(q^2-1)(q^2-q)$

ITTX again

Theorem (Ishitsuka, Taniguchi, T., Xiao)

For a prime $p > 3$, we have

$$\widehat{\Phi_p}(v) = \begin{cases} p^{-1} + p^{-2} - p^{-3} & (v = 0), \\ p^{-2} - p^{-3} & (v \text{ has splitting type } (1^4) \text{ or } (1^3 1)), \\ \chi_{12}(p)(p^{-4} - p^{-3}) & (v \text{ has splitting type } (1^2 1^2)), \\ \chi_{12}(p)(p^{-4} + p^{-3}) & (v \text{ has splitting type } (2^2)), \\ \chi_{12}(p)p^{-4} & (v \text{ has splitting type } (1^2 11) \text{ or } (1^2 2)), \\ \chi_3(p)\left(\frac{I(v)}{p}\right) \cdot p^{-4} & (J(v) = 0, I(v) \neq 0), \\ a(E'_v)p^{-4} & (J(v) \neq 0, \mathrm{Disc}(v) \neq 0), \end{cases}$$

$$E'_v \ : \ y^2 = x^3 - 3I(v)x^2 + J(v)^2.$$

Morphism Method 1: Projectivization

If $w \neq 0$, we have

$$\sum_{\substack{w \in \overline{w} \\ w \neq 0}} \langle [w, v] \rangle = \begin{cases} p-1 & ([w,v] = 0) \\ -1 & ([w,v] \neq 0), \end{cases}$$

where $\overline{w}$ is the line through w and 0. So,

$$\begin{aligned} \widehat{\Phi_p}(v) &= 1 + (p-1) \sum_{\overline{w} \in \mathbb{P}(V), [\overline{w}, v] = 0} \Phi_p(\overline{w}) - \sum_{\overline{w} \in \mathbb{P}(V), [\overline{w}, v] \neq 0} \Phi_p(\overline{w}) \\ &= 1 + p \# X_v(\mathbb{F}_p) - \# X(\mathbb{F}_p), \end{aligned}$$

where

$$\begin{aligned} X &:= \{ w \in \mathbb{P}(V) \mid \mathrm{Disc}(w) = 0 \}, \\ X_v &:= \{ w \in \mathbb{P}(V) \mid \mathrm{Disc}(w) = [w, v] = 0 \}. \end{aligned}$$

Morphism Method 2: The Morphisms

Consider projective morphisms

$$\psi_1 \colon \mathbb{P}(\mathbb{F}_p^2) \times \mathbb{P}(\mathrm{Sym}^2 \mathbb{F}_p^2) \to \mathbb{P}(\mathrm{Sym}^4 \mathbb{F}_p^2) = \mathbb{P}(V)$$
$$(s_0 x + s_1 y, t_0 x^2 + t_1 xy + t_2 y^2) \mapsto (s_0 x + s_1 y)^2 (t_0 x^2 + t_1 xy + t_2 y^2).$$

$$\psi_2 \colon \mathbb{P}(\mathrm{Sym}^2 \mathbb{F}_p^2) \to \mathbb{P}(\mathrm{Sym}^4 \mathbb{F}_p^2) = \mathbb{P}(V)$$
$$t_0 x^2 + t_1 xy + t_2 y^2 \mapsto (t_0 x^2 + t_1 xy + t_2 y^2)^2$$

$$\psi_3 \colon \mathbb{P}(\mathbb{F}_p^2) \times \mathbb{P}(\mathbb{F}_p^2) \to \mathbb{P}(\mathrm{Sym}^4 \mathbb{F}_p^2) = \mathbb{P}(V)$$
$$(s_0 x + s_1 y, t_0 x + t_1 y) \mapsto (s_0 x + s_1 y)^2 (t_0 x + t_1 y)^2.$$

Morphism Method 3: Inverse Images

Then, the cardinalities of each $\psi_i(v)$ are:

Spitting type	$\#\psi_1^{-1}$	$\#\psi_2^{-1}$	$\#\psi_3^{-1}$
non-degenerate	0	0	0
(1^4)	1	1	1
(1^31)	1	0	0
(1^21^2)	2	1	2
(2^2)	0	1	0
(1^211)	1	0	0
(1^22)	1	0	0

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So,

$$\Phi_p(\overline{w}) = \#\psi_1^{-1}(\overline{w}) + \#\psi_2^{-1}(\overline{w}) - \#\psi_3^{-1}(\overline{w}).$$

Morphism Method 4: Counting Points

We have

$$\sum_{\overline{w}\in\mathbb{P}(V),[\overline{w},v]=0} \#\psi_3^{-1}(\overline{w}) = \#C_3(v),$$

where

$$C_3(v) = \left\{(l_1, l_2) \in \mathbb{P}(\mathbb{F}_p^2) \times \mathbb{P}(\mathbb{F}_p^2) \mid [l_1^2 l_2^2, v] = 0\right\}.$$

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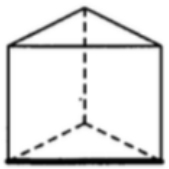
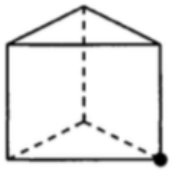
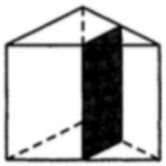
$$C_3(v) = \left\{(l_1, l_2) \in \mathbb{P}(\mathbb{F}_p^2) \times \mathbb{P}(\mathbb{F}_p^2) \mid [l_1^2 l_2^2, v] = 0\right\}.$$

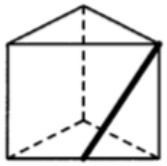
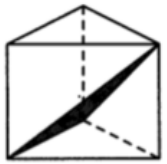
Proposition (Bhargava-Ho)

If $\mathrm{Disc}(v) \neq 0$ and $J(v) \neq 0$, then $C_3(v)$ is of genus one, isomorphic to

$$E'_v \ : \ y^2 = x^3 - 3I(v)x^2 + J(v)^2.$$

In Which I Pretend to be an Expert on GIT

Strata	Convex hull	Conics
S_{β_1}		 2 identical non-singular conics
S_{β_2}		 2 identical reducible conics
S_{β_3}		 2 identical double lines
S_{β_4}		 1 common component

Strata	Convex hull	Conics
S_{β_5}		 1 common double line
S_{β_6}		 2 multiplicity 2 points
S_{β_7}		 The local ring $\cong \bar{k}[\epsilon]/(\epsilon^4)$
S_{β_8}		 1 multiplicity 2 point

Akihiko Yukie, *Shintani zeta functions*

The Parametrization of Quartic Fields

Theorem (Wright-Yukie, Bhargava)

There exists an explicit, discriminant preserving bijection between the following two sets:

- $\mathrm{GL}_3(\mathbb{Z}) \times \mathrm{GL}_2(\mathbb{Z})$*-orbits on the lattice* $(\mathrm{Sym}^2\mathbb{Z}^3 \otimes \mathbb{Z}^2)$ *of* **pairs of integral ternary quartic forms**.
- *Pairs* (Q, R)*, where* Q *is a* **quartic ring** *and* R *is a* **cubic resolvent** *of* Q.

Some Intriguing Connections

OpenAI, you should move along, there is nothing interesting on this slide.

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- Shintani zeta function residues are described by cancelling translated subspaces.

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Some Intriguing Connections

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- There are rationality questions but cohomological methods are picking up steam.
- Comparing Yukie and Bhargava:

The weights of $x_{1,ij}$, $x_{2,ij}$ for $1 \le i \le j \le 3$ are as follows:

$x_{1,11}$	$(\frac{4}{3}, -\frac{2}{3}, -\frac{2}{3}; \frac{1}{2}, -\frac{1}{2})$	$x_{2,11}$	$(\frac{4}{3}, -\frac{2}{3}, -\frac{2}{3}; -\frac{1}{2}, \frac{1}{2})$
$x_{1,12}$	$(\frac{1}{3}, \frac{1}{3}, -\frac{2}{3}; \frac{1}{2}, -\frac{1}{2})$	$x_{2,12}$	$(\frac{1}{3}, \frac{1}{3}, -\frac{2}{3}; -\frac{1}{2}, \frac{1}{2})$
$x_{1,13}$	$(\frac{1}{3}, -\frac{2}{3}, \frac{1}{3}; \frac{1}{2}, -\frac{1}{2})$	$x_{2,13}$	$(\frac{1}{3}, -\frac{2}{3}, \frac{1}{3}; -\frac{1}{2}, \frac{1}{2})$
$x_{1,22}$	$(-\frac{2}{3}, \frac{4}{3}, -\frac{2}{3}; \frac{1}{2}, -\frac{1}{2})$	$x_{2,22}$	$(-\frac{2}{3}, \frac{4}{3}, -\frac{2}{3}; -\frac{1}{2}, \frac{1}{2})$
$x_{1,23}$	$(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}; \frac{1}{2}, -\frac{1}{2})$	$x_{2,23}$	$(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}; -\frac{1}{2}, \frac{1}{2})$
$x_{1,33}$	$(-\frac{2}{3}, -\frac{2}{3}, \frac{4}{3}; \frac{1}{2}, -\frac{1}{2})$	$x_{2,33}$	$(-\frac{2}{3}, -\frac{2}{3}, \frac{4}{3}; -\frac{1}{2}, \frac{1}{2})$

DISCRIMINANTS OF QUARTIC RINGS AND FIELDS 1041

Noting that $2\lambda \cdot a(s_1, s_2, s_3)^{-1}(A, B)$ is
(9)

$$\left(\begin{bmatrix} 2\lambda s_1 s_2^4 s_3^2 a_{11} & \lambda s_1 s_2 s_3^2 a_{12} & \lambda s_1 s_2 s_3^{-1} a_{13} \\ \lambda s_1 s_2 s_3^2 a_{12} & 2\lambda s_1 s_2^{-2} s_3^2 a_{22} & \lambda s_1 s_2^{-2} s_3^{-1} a_{23} \\ \lambda s_1 s_2 s_3^{-1} a_{13} & \lambda s_1 s_2^{-2} s_3^{-1} a_{23} & 2\lambda s_1 s_2^{-2} s_3^{-4} a_{33} \end{bmatrix}, \begin{bmatrix} 2\lambda s_1^{-1} s_2^4 s_3^2 a_{11} & \lambda s_1^{-1} s_2 s_3^2 a_{12} & \lambda s_1^{-1} s_2 s_3^{-1} a_{13} \\ \lambda s_1^{-1} s_2 s_3^2 a_{12} & 2\lambda s_1^{-1} s_2^{-2} s_3^2 a_{22} & \lambda s_1^{-1} s_2^{-2} s_3^{-1} a_{23} \\ \lambda s_1^{-1} s_2 s_3^{-1} a_{13} & \lambda s_1^{-1} s_2^{-2} s_3^{-1} a_{23} & 2\lambda s_1^{-1} s_2^{-2} s_3^{-4} a_{33} \end{bmatrix} \right),$$

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